

(i) SMILES

CCC(C)(C)[C@H](NC(C)=O)C(=O)NC, CNC(=O)C[C@@H](C)NC(C)=O,

CNC(=O)C[C@@H]1CCN(C(C)=O)C1,

CNC(=O)C[C@H](CCCC[NH3+])NC(C)=O,

CNC(=O)[C@@H](CCC(C)C)NC(C)=O,

192 characters | No semantic encoding | Not human-readable

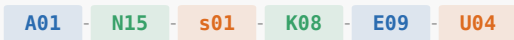
CNC(=O)[C@@H]1C[C@H]1[C@@H](NC(C)=O)c1ccccc1

(ii) HELM

PEPTIDE1{[A01].[N15].[s01].[K08].[E09].[U04]}\$\$\$\$V2.0

53 characters | Monomer IDs only | Requires external library

(iii) ResToken^(ours)



alpha

L-config
large

beta

D-config
small

gamma

L-config
small

beta

L-config
large

alpha

D-config
large

gamma

D-config
large

Token encoding scheme:

Prefix letter → backbone type

A-E = alpha

K-O = beta

U-Y = gamma

Case → chirality

UPPER (A01) = L-config

lower (s01) = D-config

Suffix digits → side-chain ID

● alpha
● beta
● gamma

23 characters | Semantically encoded | Human-readable